INCH-POUND

MIL-PRF-32152(SH)
11 April 2005

PERFORMANCE SPECIFICATION

CLOSURE, HVAC, BUTTERFLY TYPE, SHIPBOARD SERVICE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes watertight, fire resistant butterfly closures used in heating ventilation and air conditioning (HVAC) systems onboard Naval surface ships.

1.2 Classification. Closures are of the following styles, types, sizes, and structures as specified (see 6.2):

1.2.1 Styles. The styles of butterfly closures are as follows:

Style R - Round

Style K - Flat Oval

1.2.2 Types. The types of butterfly closures are as follows:

Type I - Handle operated

Type II - Handwheel operated

Type III - Motor operated

1.2.3 Sizes. The sizes of butterfly closures are as follows:

Style R nominal size in inches		Style K nominal size in inches
2½	12	31 (7x3)
3	14	32 (8x4)
3½	16	33 (9x5)
4	18	34 (11x6)
5	20	35 (13x7)
6	24	36 (16x8)
8	30	37 (20x9)
10	36	38 (24x10)
		39 (24x10)
		40 (30x16)
		41 (30x20)

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navsea.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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1.2.4 Structure. The structure code of butterfly closures is as follows:

Nonmagnetic structure (NM) or the option to be not applicable (NA).

1.3 Part or identifying number (PIN). PINs to be used for watertight, fire resistant butterfly closures acquired to this specification are created as follows:

M	**32152**	**-**	**X**	**X**	**XXX**	**XX**
Prefix for Military Specification	Specification Number		Style (see code below)	Type (see code below)	Size (see code below)	Structure (see code below)

Style Code		Type Code		Size Code		Structure Code	
Style	Code	Type	Code	Size	Code [1/]	Structure	Code
R	R	I	1	**Style R only**		Nonmagnetic	NM
K	K	II	2	2½	02H	Not applicable	NA
		III	3	3	03W		
				3½	03H		
				4	04W		
				5	05W		
				6	06W		
				8	08W		
				10	10W		
				12	12W		
				14	14W		
				16	16W		
				18	18W		
				20	20W		
				24	24W		
				30	30W		
				36	36W		
				Style K only			
				31 (7x3)	31W		
				32 (8x4)	32W		
				33 (9x5)	33W		
				34 (11x6)	34W		
				35 (13x7)	35W		
				36 (16x8)	36W		
				37 (20x9)	37W		
				38 (24x10)	38W		
				39 (24x10)	39W		
				40 (30x16)	40W		
				41 (30x20)	41W		

NOTE:

[1/] "W" denotes a whole number; "H" denotes the fraction "½".

Examples: M32152-R202HNM

M32152-K338WNA

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

DOD-V-24657	-	Valve Actuator, Direct Coupled, Gear Driven Electrically Powered for Naval Shipboard Propulsion and Auxiliary Systems (Metric)
MIL-S-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements For

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I–Environmental and Type II – Internally Excited)
MIL-STD-798	-	Nondestructive Testing, Welding, Quality Control, Material Control and Identification and HI-Shock Test Requirements for Piping System Components for Naval Shipboard Use
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1399-300	-	Interface Standard for Shipboard Systems, Electric Power, Alternating Current (Metric)
MIL-STD-2142	-	Magnetic Silencing Characteristics, Measurement Of (Metric)

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVSEA TECHNICAL PUBLICATION

S9074-AR-GIB-010/278	-	Requirements for Fabrication, Welding and Inspection, Casting Inspection and Repair for Machinery, Piping and Pressure Vessels

(Copies of this document are available from the Naval Inventory Control Point, 700 Robbins Avenue, Attn: Code 0862 (Cash Sale), Philadelphia, PA 19111 or online at www.nll.navsup.navy.mil.)

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NAVSEA DRAWINGS

NAVSEA 803-1385620	-	Handwheels for Valves
NAVSEA 804-1749102	-	Ventilation Closure Watertight and Fire-Resistant Model “R” Round
NAVSEA 804-1749103	-	Ventilation Closure Watertight and Fire Resistant Model “K” Flat Oval

(Copies of these drawings are available from the Naval Inventory Control Point, 700 Robbins Avenue, Attn: Code 0862 (Cash Sale), Philadelphia, PA 19111 or online at www.nll.navsup.navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR MOVEMENT AND CONTROL ASSOCIATION (AMCA) INTERNATIONAL, INC.

AMCA 500	-	Test Methods for Louvers, Dampers, and Shutters (DoD adopted)

(Copies of this document are available from the Air Movement and Control Association, Inc., 30 West University Drive, Arlington Heights, IL 60004 or online at www.amca.org.)

AMERICAN WELDING SOCIETY (AWS)

B2.1	-	Welding Procedure and Performance Qualifications, Standard for (DoD adopted)
B2.2	-	Brazing Procedure and Performance Qualification, Standard for (DoD adopted)

(Copies of these documents are available from the American Welding Society, 550 NW LeJeune Road, Miami, FL 33216 or online at www.aws.org.)

ASME INTERNATIONAL

ASME B1.1	-	Unified Inch Screw Threads, UN and UNR Thread Form (DoD adopted)
ASME B18.2.1	-	Square and Hex Bolts and Screws, Inch Series (DoD adopted)

(Copies of these documents are available from ASME International, 22 Law Drive, P.O. Box 2900, Fairfield, NJ 07007-2900 or online at www.asme.org.)

ASTM INTERNATIONAL

ASTM F1166	-	Standard Practice for Human Engineering Design for Marine Systems, Equipment and Facilities (DoD adopted)

(Copies of this document are available from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials which are not inherently corrosion resistant are prohibited. Cast iron shall not be used. Material degraded during the fabrication process shall be normalized to restore corrosion-resisting properties before being assembled in the closure. Selected materials shall be capable of meeting all of the operational and environmental requirements specified herein.

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3.2.1 Hazardous material. Materials for use in the construction of closures shall have no effect on the health of personnel when the materials are used for intended purpose. Regardless of other requirements, materials and parts containing asbestos, cadmium, lithium, mercury, or radioactive material shall not be used.

3.2.2 Fasteners. Material for bolts, nuts, screws, and similar fasteners that are not inherently corrosion resistant are prohibited. Self-tapping sheet metal screws shall not be used.

3.2.3 Dissimilar metals. The closure shall not be degraded due to electrolysis in accordance with MIL-STD-889.

3.2.4 Nonmagnetic material. When nonmagnetic closures are specified (see 6.2), parts shall be fabricated of nonmagnetic material which has a maximum relative permeability of less than 2.0 after fabrication.

3.2.5 Gasket material. Seals shall be heat and moisture resistant. Recommend primary seals be elastomer and secondary seals be intumescent.

3.2.6 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Painting. Closure and component items shall be painted if specified (see 6.2).

3.4 Welding and allied processes. Surfaces of parts to be welded or brazed shall be free from rust, scale, paint, grease, and other foreign matter. Welding and allied processes shall be performed by personnel certified to AWS B2.1 and B2.2. NAVSEA Technical Publication S9074-AR-GIB-010/278 shall be used for guidance.

3.5 Identification plates. Each closure shall be provided with a permanent corrosion-resistant identification plate. The method of attachment for the identification plate shall be permanent and corrosion-resistant. The plate shall contain the following information:

a. National stock number (NSN)
b. Manufacturers name, commercial and government entity (CAGE), and part number
c. PIN code
d. Contract or order number
e. Preferred flow direction arrow

3.6 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable with regard to function, performance, and strength.

3.7 Workmanship. The closure shall be free from defects that affect appearance and operation. Sharp edges, burrs, and other imperfections shall be removed from all parts subject to contact with personnel to prevent cuts during repair and maintenance. Closure edges shall be square and fasteners shall be in place and secure.

3.8 Human engineering. ASTM F1166 shall be utilized as a guideline in applying human engineering design criteria for the closure.

3.9 Maintainability. The closure shall be constructed so that:

a. Disc seals are replaceable without removing stem-disc assembly.
b. Disassembly can be accomplished using commonly available tools and without removal of closure body from ductwork.

3.10 Standardization requirements.

3.10.1 Sizes. Sizes of Style R closures and dimensions necessary to ensure interchangeability between manufacturers are shown on Figure 1. Sizes of Style K closures and dimensions to ensure interchangeability between manufacturers are shown on Figure 2. Examples of butterfly closures previously supplied to the U.S. Navy are shown in NAVSEA drawings 804-1749102 and 804-1749103.

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3.10.2 Closure, operators. All closures shall be designed to permit the use of manual handwheel or electrical type operators. In addition, sizes R2½ through R6 and sizes K31 through K34 shall also permit the use of handle operators. Maximum allowable dimensions are shown on Figure 3.

3.10.3 Closure body. The closure body shall be designed for installation between bolted duct flanges without the use of additional flange gaskets. Closure body seals shall provide for watertightness of flange to closure connection under shock, vibration, and fire. Seals shall be replaceable.

3.10.3.1 Alignment and retention. Means shall be provided for aligning and retaining the closure between companion duct flanges. An acceptable method is shown on Figure 1 for Style R closures and Figure 2 for Style K closures.

3.10.4 Disc. Closures shall be designed for adequate disc to duct clearance for the inside duct diameter "A" to ensure there is no interference between the disc and adjacent components.

3.10.5 Duct to closure flange. Duct to closure flange requirements are indicated in Figures 4 and 5.

3.10.6 Operator rotation. Clockwise rotation of handwheels and handles shall be designed to close the closure.

3.11 Design requirements. Each closure assembly shall consist of closure body, disc, stems, gaskets, packing, bushings and or bearings and seals and mounting surface for operators.

3.11.1 Seats. The sealing mechanism shall be part of the disc. The disc and sealing mechanism shall provide closure watertightness under shock, vibration, and fire. Seats shall be replaceable.

3.11.2 Seal. The disc seal shall be replaceable without removing the stem-disc assembly and shall be reversible or designed to physically preclude improper installation.

3.11.3 Position indicator. All closures shall be provided with a local position indicator.

3.12 Performance requirements.

3.12.1 Stem bushings. Stem bushings shall be non-seizing after exposure to a fire.

3.12.2 Stem seals. Stem seals shall be watertight and fire-resistant.

3.12.3 Overtravel. Closure design shall prevent disc overtravel to ensure positive watertight seating of disc.

3.12.4 Electrical operator. Electrical operator conforming to DOD-V-24657 shall be provided with a manual handwheel override and shall be manually operable before and after a 30-minute fire. (see Figure A-1)

3.12.5 Flow performance. Maximum pressure loss for air at 70 °F and 2500 fpm flowing through the closure shall be 0.37 inches water for Style R closure in a duct and 0.13 inches water for Style K closure in duct.

3.12.6 Closing torque. Closures shall be capable of being closed by an electrical operator, or by the clockwise movement of a handle operator, or by the handwheel of an operator. Closures shall seat in accordance with 3.12.8 with a maximum pull of 60 pounds force on the rim of the handwheel (NAVSEA drawing 803-1385620) or at a point 2 inches less than the length of the handle (effective length). The handwheel/handle shall be capable of providing a minimum of two times the normal operating torque.

3.12.7 Operating life. The closure seal shall be maintained through testing of 2000 cycles at 15 PSIG with air in accordance with 4.5.3.

3.12.8 Closure seat. The closure seat shall provide a bubble tight seal when subjected to air pressure of 30 PSIG in the recommended pressure seal direction including after the completion of the shock and vibration tests.

3.12.9 Closure body. Closure body shall not leak from pressure boundaries when tested as specified in 4.5.8.

3.12.10 Fire resistance. The closure and operators shall be manually operable after exposure to a fire of 1500 °F for a minimum of 30 minutes and after the valve body attains a minimum temperature of 800 ºF.

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3.13 Environmental conditions.

3.13.1 General shipboard design conditions. General shipboard design characteristics shall be as follows:

a. Power source quality in accordance with MIL-STD-1399-300.

b. Perform in accordance with requirements herein in a range of ambient temperature between 0 °F and 200 °F.

c. Perform in accordance with the requirements herein at a 15 PSIG normal working pressure and a 30 PSIG maximum working pressure with a flow media of air.

3.13.2 Vibration. Closure and operator assembly shall be designed such that no damage or malfunction be caused by environmental vibrations in accordance with MIL-STD-167-1.

3.13.3 High impact shock resistance. Closure and operator assembly shall satisfy the requirements of MIL-S-901 for Grade A, Class I equipment.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on each style, type, size, and structure of closure and operator assembly when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.5.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of Table I.

TABLE I. Test agenda.

Tests	Requirement	Test	First Article	Conformance
Pressure loss	3.12.5	4.5.1	All [1/]	
Torque	3.12.6	4.5.2	All	All
Operating	3.12.7	4.5.3	All	
Permeability	3.2.4	4.5.4	All (only nonmagnetic)	All (only nonmagnetic)
Vibration	3.13.2	4.5.5	All [2/]	
Shock	3.13.3	4.5.6	All [2/]	
Seat leakage	3.12.8	4.5.7	All	All
Hydrostatic	3.12.9	4.5.8	All	All
Fire resistance	3.12.4, 3.12.10	4.5.9	All [2/]	
Life maintainability	3.9	4.5.10	1st production unit	

NOTES:

[1/] Pressure loss for valves smaller than valve tested may be extrapolated provided the flow path geometry is a scaled version of the test valve.

[2/] Tests of a closure and operator size for vibration, shock, and fire resistance tests qualifies the size tested and all sizes below the size tested for these tests provided the design and materials of construction are similar.

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4.4 Examination. Closures shall be examined and measured to verify compliance with the requirements of this specification not involving tests. Examination shall be conducted as specified in 4.4.1, 4.4.2, and Table II. Any closure in the sample containing one or more defects shall not be offered for delivery.

4.4.1 Lot. All closures of the same size and type offered for delivery at one time shall be considered a lot.

4.4.2 Sampling for examination. A random sampling of closures shall be selected for each lot in accordance with Table III for the examination specified in 4.4.

TABLE II. Classification of defects.

Categories	Defects
Critical	
1	None defined.
Major	
101	Size not as specified.
102	Incomplete, component parts missing (flange seal, seat seal, stem bushings, handwheel, handle, electric actuator).
103	Materials defective or not as specified.
104	Limited dimensions exceeded.
105	Flange holes and mounting dimensions not as specified.
106	Welding incomplete, not free of cracks, non-fusion, heavy porosity, heavy undercut, slag inclusions.
107	Sharp edges and burrs not removed from parts subject to personnel contact such as handwheels and handles.
108	Gaskets/seals, missing or defective.
109	Bolts, nuts, and screws not tight, missing.
110	Splices in wire.
111	Electrical connections, not at terminal of components or terminal blocks.
112	Wiring not neat, tied, or clamped.
113	Closure does not close when handwheel/handle is turned in the clockwise direction.
114	Closure requires excess force to operate.
115	Position indicator not provided.
116	Marking, identification plate not complete, missing not permanent, illegible, or not as specified.
117	Wiring diagram missing (electric actuator).
118	Painting (as applicable) not as specified.

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TABLE III. Sampling for examination.

Lot size number of closures	Sample size number of closures
1 to 6	All
7 to 15	7
16 to 25	10
26 to 40	13
41 to 65	17
66 to 110	22
111 to 180	28
181 to 300	35
301 to 500	45

4.5 Tests. Tests shall be conducted in accordance with 4.5.1 through 4.6.

4.5.1 Flow performance. Pressure drop test shall be conducted per AMCA 500 and shall conform to the requirements of 3.12.5. Losses through smaller closures may be extrapolated provided the flow path geometry is a scaled version of the test valve.

4.5.2 Closing torque test. Conduct tests in accordance with AMCA 500 to verify that operating torque meets the requirements of 3.12.6. Testing shall be conducted at zero airflow and at zero pressure differential in the closed position.

4.5.3 Operating life test. The closure shall maintain a bubble tight seal when tested in accordance with AMCA 500 and shall conform to the requirements of 3.12.7. Wearing parts may be replaced after accomplishment of operating life test.

4.5.4 Permeability tests. A permeability test of nonmagnetic material used in construction of the closure shall be conducted in accordance with MIL-STD-2142, Test 501.

4.5.5 Vibration tests. The closure and operator assembly shall be subjected to Type I vibration tests in accordance with MIL-STD-167-1. The exploratory test specified in MIL-STD-167-1 shall include frequencies up to 33 Hz at the table amplitude specified therein. The vibration test shall be conducted as specified in 4.6 and Table I. Any unit that fails to meet any of the requirements shall be rejected.

4.5.6 High impact shock tests. The closure and operator assembly shall be shock tested for Grade A, Class I shock of MIL-S-901. MIL-STD-798 shall be used as guidance. The closure shall be tested in the full open (normal) position and full closed positions and subject to the normal working pressure of 15 PSIG at ambient temperature (in recommended pressure seal direction). Evidence of fragmentation, deformation that will cause active interference between parts, or failure to operate shall be cause for rejection.

4.5.7 Seat leakage test. Testing for leakage shall be in accordance with AMCA 500 using the airflow leakage rate test for a zero leakage closure. The torque applied shall be in accordance with 3.12.6. This test shall be performed after the shock and vibration tests and when specified in Table I.

4.5.8 Hydrostatic test. The closure shall be pressurized with water to 45 PSI and shall meet the leakage requirements as specified in 3.12.9.

4.5.9 Fire resistance test. The closure and operator assembly shall meet the test requirements specified in Appendix A and Figure A-1.

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4.5.10 Maintainability demonstration. The first production unit shall be examined after testing, and the capability to meet the requirements of 3.9 to maintain, disassemble, and repair the unit shall be demonstrated. The demonstration shall be conducted utilizing the recommended tools and with other than expert mechanics. Evidence that maintainability of the closure cannot be accomplished by other than expert mechanics shall be cause for failure of the demonstration. The maintainability demonstration shall include but not be limited to the following:

a. Replacement of disc seal.
b. Removal of disc without removal of closure body from ductwork.
c. Replacement of closure body seal.

4.6 Test schedule. The schedule for testing the closure and operator assembly shall be performed as shown in Table I.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The closure operator assembly is intended to be used as a shutoff device to prevent the spread of fire, smoke, and flooding via HVAC ductwork from a watertight subdivision, compartment, or deck to another watertight subdivision, compartment, or deck. The closure is also intended to survive a 1500 °F fire and be capable of manual operation. It is intended that the closure, operator, and operator fasteners be provided as a package to ensure proper mounting and operation under conditions set forth in 3 and 4.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Style, type, size, and structure required (see 1.2).
c. When first article is required (see 3.1).
d. Whether nonmagnetic closures are required (see 3.2.4).
e. Paint requirements, if required (see 3.3).
f. Packaging requirements (see 5.1).
g. Whether fire test records are required (see Appendix A).

6.3 Subject term (key word) listing.

Flat oval closure
Operator
Round closure
Shutoff device
Ventilation

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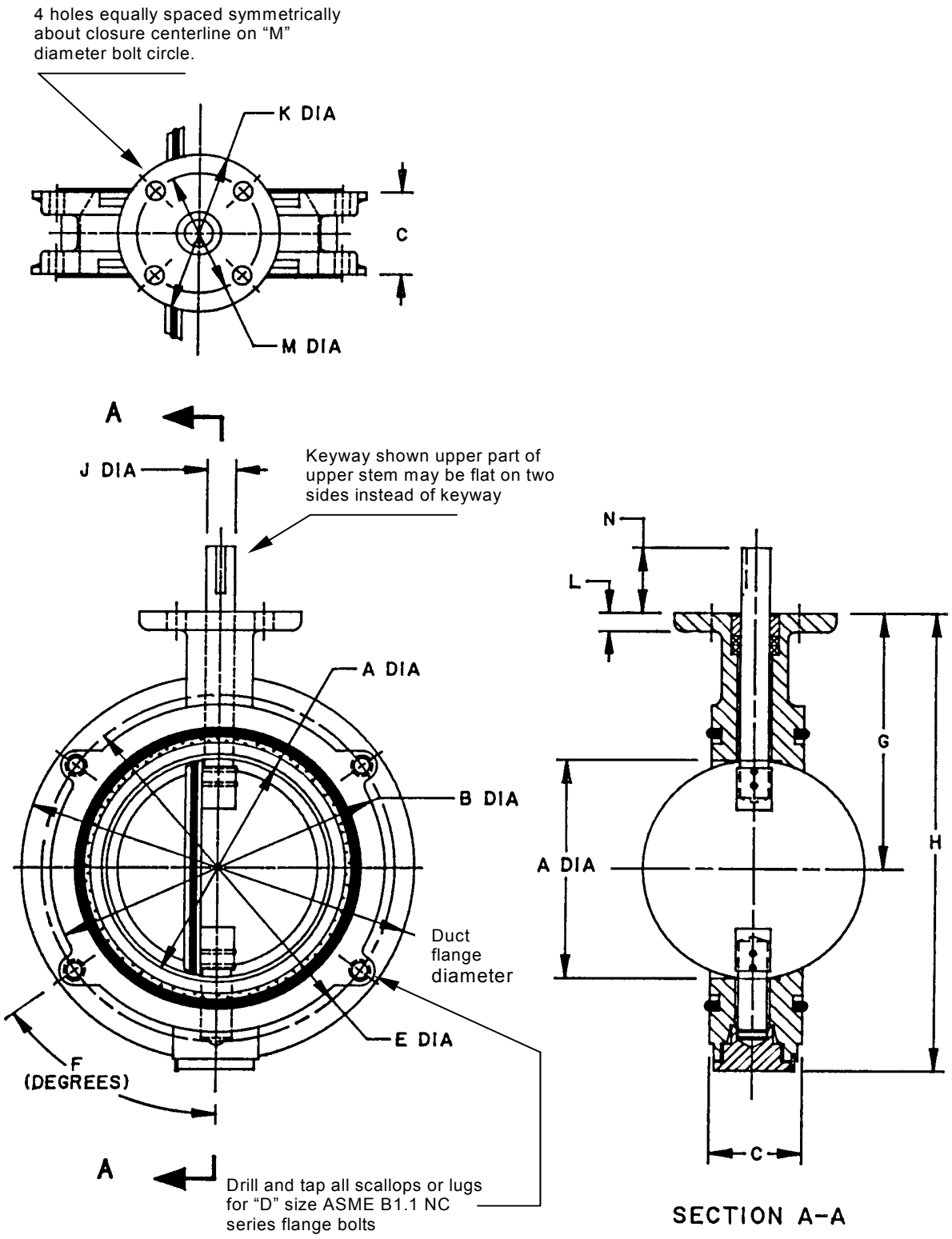


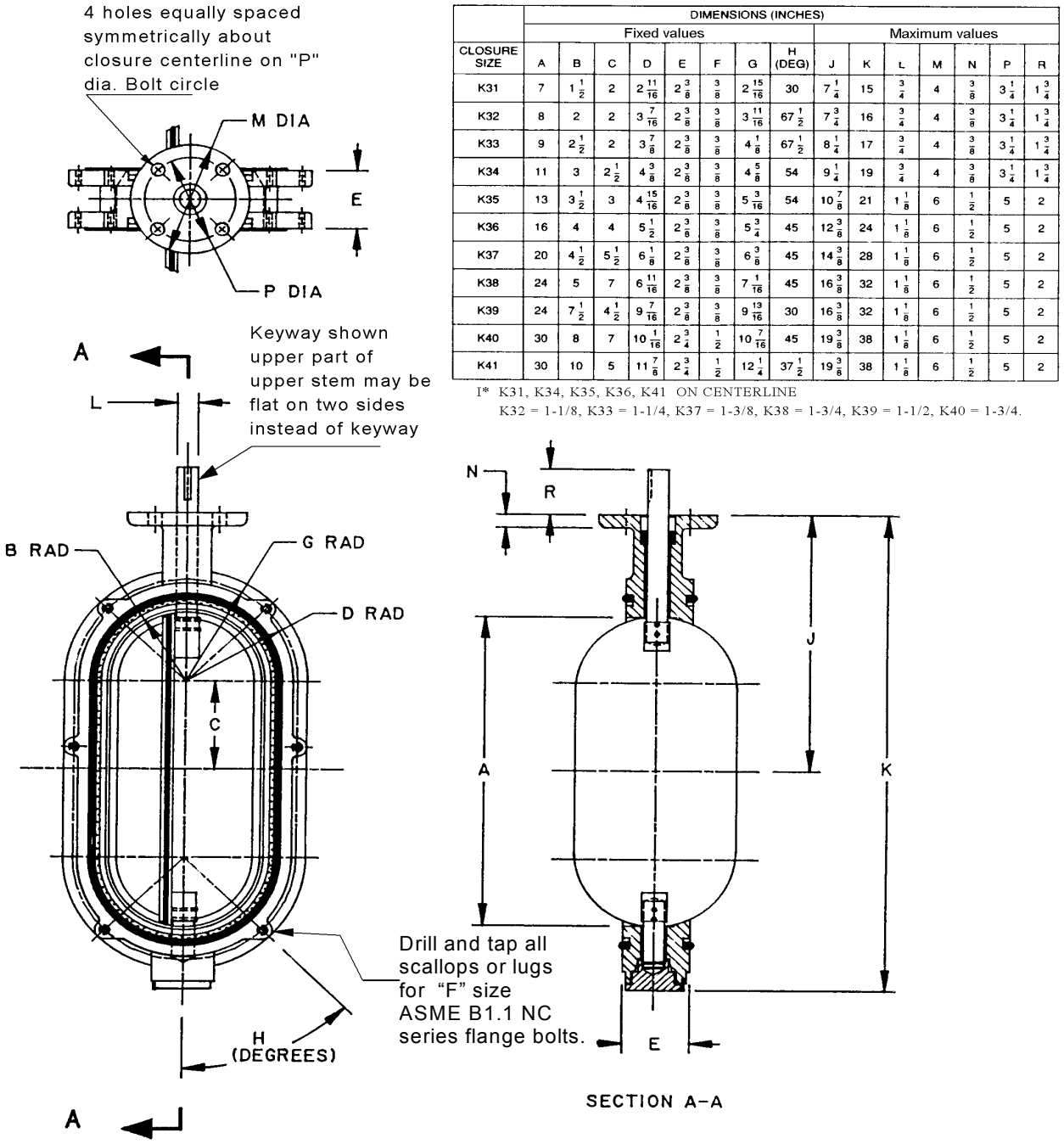
FIGURE 1. Style R closure.

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CLOSURE SIZE	DIMENSIONS (INCHES)												
	Fixed Values						Maximum Values						
	A	B	C	D (DIA)	E	F (DEG)	G	H	J	K	L	M	N
R2 ½	2.47	$4\frac{7}{8}$	$1\frac{3}{4}$	$\frac{3}{8}$	$5\frac{3}{8}$	30	5	10	$\frac{3}{4}$	$4\frac{1}{4}$	$\frac{1}{2}$	$3\frac{1}{4}$	$1\frac{3}{4}$
R3	3.07	$5\frac{3}{8}$	$1\frac{3}{4}$	$\frac{3}{8}$	$5\frac{7}{8}$	30	$5\frac{1}{4}$	$10\frac{1}{2}$	$\frac{3}{4}$	$4\frac{1}{4}$	$\frac{1}{2}$	$3\frac{1}{4}$	$1\frac{3}{4}$
R3 ½	3.55	$6\frac{3}{8}$	$1\frac{7}{8}$	$\frac{3}{8}$	$6\frac{7}{8}$	$67\frac{1}{2}$	$5\frac{1}{2}$	11	$\frac{3}{4}$	$4\frac{1}{4}$	$\frac{1}{2}$	$3\frac{1}{4}$	$1\frac{3}{4}$
R4	4.03	$6\frac{7}{8}$	2	$\frac{3}{8}$	$7\frac{3}{8}$	$67\frac{1}{2}$	6	12	$\frac{3}{4}$	$4\frac{1}{4}$	$\frac{1}{2}$	$3\frac{1}{4}$	$1\frac{3}{4}$
R5	5.05	$7\frac{3}{4}$	$2\frac{1}{8}$	$\frac{3}{8}$	$8\frac{1}{4}$	$67\frac{1}{2}$	$6\frac{1}{2}$	13	$\frac{3}{4}$	$4\frac{1}{4}$	$\frac{1}{2}$	$3\frac{1}{4}$	$1\frac{3}{4}$
R6	6.07	$8\frac{3}{4}$	$2\frac{1}{8}$	$\frac{3}{8}$	$9\frac{1}{4}$	54	$6\frac{7}{8}$	14	$\frac{3}{4}$	$4\frac{1}{4}$	$\frac{1}{2}$	$3\frac{1}{4}$	$1\frac{3}{4}$
R8	7.98	11	$2\frac{3}{8}$	$\frac{3}{8}$	$11\frac{1}{2}$	45	$8\frac{5}{8}$	16	$1\frac{1}{8}$	6	$\frac{1}{2}$	5	2
R10	10.02	$13\frac{3}{8}$	$2\frac{9}{16}$	$\frac{3}{8}$	$13\frac{7}{8}$	50	$9\frac{3}{4}$	18	$1\frac{1}{8}$	6	$\frac{1}{2}$	5	2
R12	12.00	$16\frac{1}{8}$	$3\frac{1}{16}$	$\frac{3}{8}$	$16\frac{5}{8}$	50	$10\frac{3}{4}$	20	$1\frac{1}{8}$	6	$\frac{1}{2}$	5	2
R14	14.00	$17\frac{3}{4}$	$2\frac{3}{4}$	$\frac{3}{8}$	$18\frac{1}{4}$	45	13	22	$1\frac{1}{8}$	6	$\frac{1}{2}$	5	2
R16	16.00	$20\frac{1}{8}$	$2\frac{3}{4}$	$\frac{1}{2}$	$20\frac{7}{8}$	45	15	24	$1\frac{1}{4}$	6	$\frac{1}{2}$	5	2
R18	18.00	$21\frac{1}{2}$	$2\frac{3}{4}$	$\frac{1}{2}$	$22\frac{1}{4}$	45	16	26	$1\frac{1}{4}$	8	$\frac{5}{8}$	5	2
R20	20.00	$23\frac{3}{4}$	$2\frac{3}{4}$	$\frac{1}{2}$	$24\frac{1}{2}$	$37\frac{1}{2}$	18	28	$1\frac{1}{4}$	8	$\frac{5}{8}$	5	2
R24	24.00	$28\frac{1}{8}$	$2\frac{3}{4}$	$\frac{1}{2}$	$28\frac{7}{8}$	42	20	32	$1\frac{1}{4}$	8	$\frac{5}{8}$	5	2
R30	30.00	$34\frac{1}{2}$	$2\frac{3}{4}$	$\frac{9}{16}$	$35\frac{1}{4}$	45	21	38	$1\frac{1}{2}$	8	$\frac{3}{4}$	5	2
R36	36.00	$40\frac{3}{8}$	$4\frac{3}{4}$	$\frac{9}{16}$	$41\frac{1}{4}$	45	25	47	$1\frac{7}{8}$	8	$\frac{3}{4}$	$6\frac{1}{2}$	$4\frac{1}{4}$

FIGURE 1. Style R closure - Continued.

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CLOSURE SIZE	DIMENSIONS (INCHES)														
	Fixed values								Maximum values						
	A	B	C	D	E	F	G	H (DEG)	J	K	L	M	N	P	R
K31	7	1 1/2	2	2 11/16	2 3/8	3/8	2 15/16	30	7 1/4	15	3/4	4	3/8	3 1/4	1 3/4
K32	8	2	2	3 7/16	2 3/8	3/8	3 11/16	67 1/2	7 3/4	16	3/4	4	3/8	3 1/4	1 3/4
K33	9	2 1/2	2	3 7/8	2 3/8	3/8	4 1/8	67 1/2	8 1/4	17	3/4	4	3/8	3 1/4	1 3/4
K34	11	3	2 1/2	4 3/8	2 3/8	3/8	4 5/8	54	9 1/4	19	3/4	4	3/8	3 1/4	1 3/4
K35	13	3 1/2	3	4 15/16	2 3/8	3/8	5 3/16	54	10 7/8	21	1 1/8	6	1/2	5	2
K36	16	4	4	5 1/2	2 3/8	3/8	5 3/4	45	12 3/8	24	1 1/8	6	1/2	5	2
K37	20	4 1/2	5 1/2	6 1/8	2 3/8	3/8	6 3/8	45	14 3/8	28	1 1/8	6	1/2	5	2
K38	24	5	7	6 11/16	2 3/8	3/8	7 1/16	45	16 3/8	32	1 1/8	6	1/2	5	2
K39	24	7 1/2	4 1/2	9 7/16	2 3/8	3/8	9 13/16	30	16 3/8	32	1 1/8	6	1/2	5	2
K40	30	8	7	10 1/16	2 3/4	1/2	10 7/16	45	19 3/8	38	1 1/8	6	1/2	5	2
K41	30	10	5	11 7/8	2 3/4	1/2	12 1/4	37 1/2	19 3/8	38	1 1/8	6	1/2	5	2

1* K31, K34, K35, K36, K41 ON CENTERLINE
K32 = 1-1/8, K33 = 1-1/4, K37 = 1-3/8, K38 = 1-3/4, K39 = 1-1/2, K40 = 1-3/4.

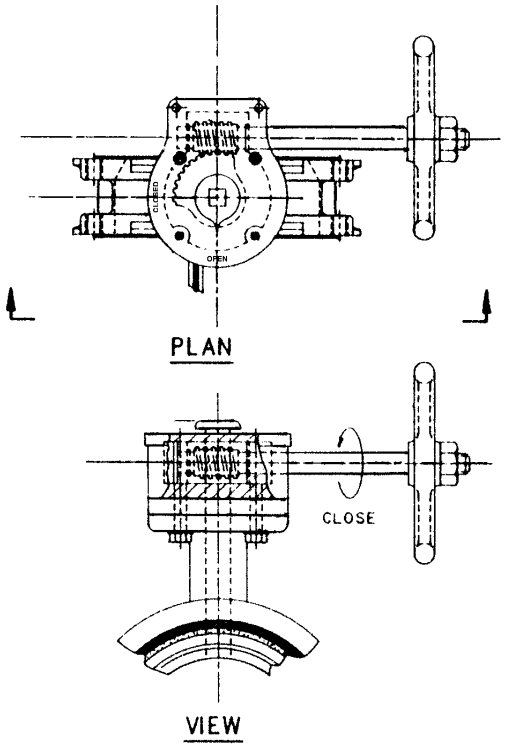
FIGURE 2. Style K closure.

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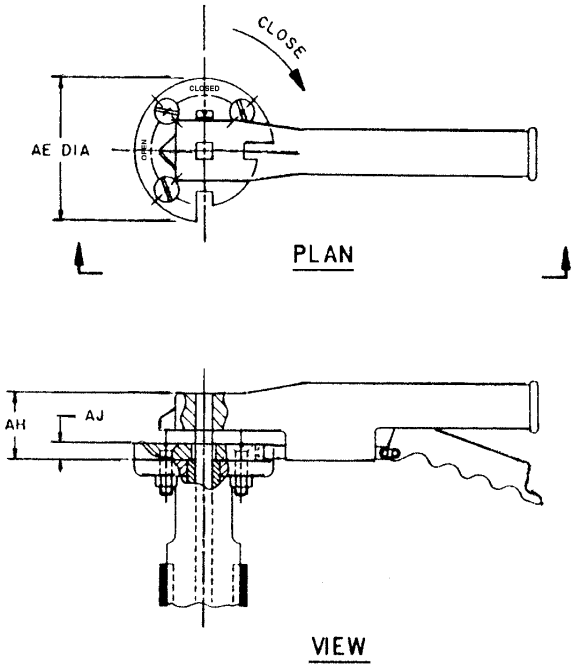
CLOSURE SIZE	DIMENSIONS (INCHES)															
	Fixed Values									Maximum Values						
	A	B	C	D	E	F	G	H (DEG)	L	J	K	L	M	N	P	R
K31	7	1 1/2	2	2 11/16	2 3/8	3/8	2 15/16	30	℄	7 1/4	15	3/4	4	3/8	3 1/4	1 3/4
K32	8	2	2	3 7/16	2 3/8	3/8	3 11/16	67 1/2	1 1/8	7 3/4	16	3/4	4	3/8	3 1/4	1 3/4
K33	9	2 1/2	2	3 7/8	2 3/8	3/8	4 1/8	67 1/2	1 1/4	8 1/4	17	3/4	4	3/8	3 1/4	1 3/4
K34	11	3	2 1/2	4 3/8	2 3/8	3/8	4 5/8	54	℄	9 1/4	19	3/4	4	3/8	3 1/4	1 3/4
K35	13	3 1/2	3	4 15/16	2 3/8	3/8	5 3/16	54	℄	10 7/8	21	1 1/8	6	1/2	5	2
K36	16	4	4	5 1/2	2 3/8	1/2	5 3/4	45	℄	12 3/8	24	1 1/8	6	1/2	5	2
K37	20	4 1/2	5 1/2	6 1/8	2 3/8	1/2	6 3/8	45	1 3/8	14 3/8	28	1 1/8	6	1/2	5	2
K38	24	5	7	6 11/16	2 3/8	1/2	7 1/16	45	1 3/4	16 3/8	32	1 1/8	6	1/2	5	2
K39	24	7 1/2	4 1/2	9 7/16	2 3/8	1/2	9 13/16	30	1 1/2	16 3/8	32	1 1/8	6	1/2	5	2
K40	30	8	7	10 1/16	2 3/4	1/2	10 7/16	45	1 3/4	19 3/8	38	1 1/8	6	1/2	5	2
K41	30	10	5	11 7/8	2 3/4	1/2	12 1/4	37 1/2	℄	19 3/8	38	1 1/8	6	1/2	5	2

FIGURE 2. Style K closure - Continued.

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TYPICAL WORM GEAR OPERATOR



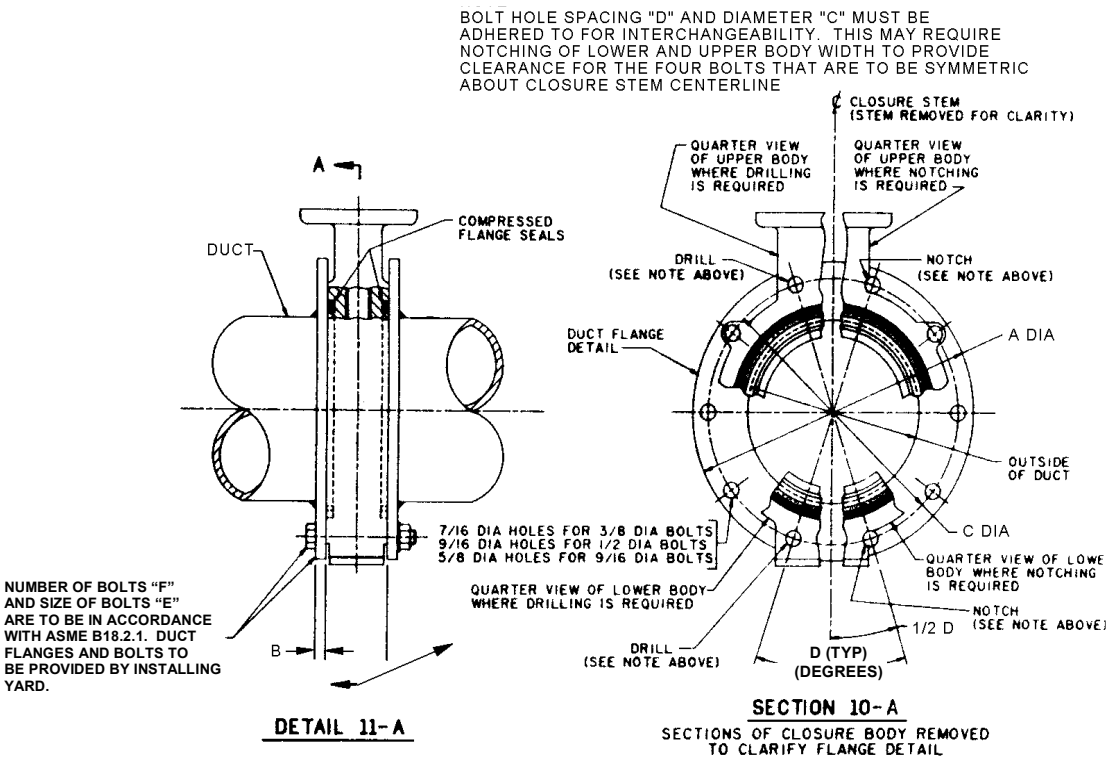
TYPICAL HANDLE OPERATOR

STYLE K CLOSURE			
Closure Maximum Dimension			
Size	AE	AH	AJ
K31	4	1¾	½
K32	4	1¾	½
K33	4	1¾	½
K34	4	1¾	½
K35	6	2	⅝
K36	6	2	⅝
K37	6	2	⅝
K38	6	2	⅝
K39	6	2	⅝
K40	6	2	⅝
K41	6	2	⅝

STYLE R CLOSURE			
Closure Maximum Dimension			
Size	AE	AH	AJ
R2½	4¼	1¾	½
R3	4¼	1¾	½
R3½	4¼	1¾	½
R4	4¼	1¾	½
R5	4¼	1¾	½
R6	4¼	1¾	½
R8	6	5	⅝
R10	6	5	⅝
R12	6	5	⅝
R14	6	5	⅝
R16	6	5	⅝
R18	8	5	⅝
R20	8	5	⅝
R24	8	5	⅝
R30	8	5	⅝
R36	8	½	¾

FIGURE 3. Closure manual operators.

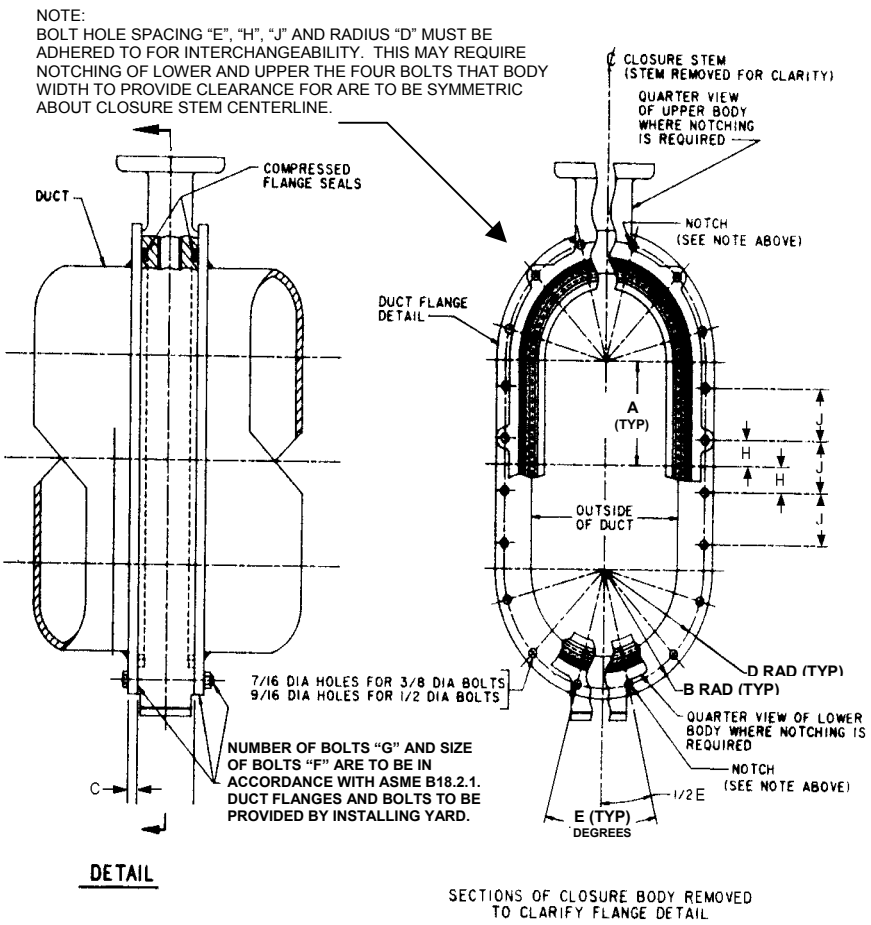
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CLOSURE SIZE	DIMENSIONS					
	A (Inch)	B (Inch)	C (Inch)	D (Deg)	E-Bolt Dia. (Inch)	F (No.)
R2½	6½	¼	5⅜	60	⅜	6
R3	7	¼	5⅞	60	⅜	6
R3½	8	¼	6⅞	45	⅜	8
R4	8½	¼	7⅜	45	⅜	8
R5	9⅜	¼	8¼	45	⅜	8
R6	10⅜	¼	9¼	36	⅜	10
R8	12⅝	⅜	11½	30	⅜	12
R10	15	⅜	13⅞	20	⅜	18
R12	17¾	⅜	16⅝	20	⅜	18
R14	19⅜	⅜	18¼	18	⅜	20
R16	22	½	20⅞	18	½	20
R18	23⅜	½	22¼	18	½	20
R20	25⅜	½	24½	15	½	24
R24	30	½	28⅞	12	½	30
R30	36¾	½	35¼	10	9/16	36
R36	43⅛	½	41¼	10	9/16	36

FIGURE 4. Typical R closure installation to duct flange.

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CLOSURE SIZE	DIMENSIONS										
	A (Inch)	B (Inch)	C (Inch)	D (Inch)	E (Deg)	No. of E Spaces	F (Inch)	G No. of Bolts	H (1)	J (Inch)	No. of J Spaces
K31	2	3½	⅜	$2^{15}/_{16}$	60	6	⅜	10	0	2	4
K32	2	4¼	⅜	$3^{11}/_{16}$	45	6	⅜	12	1⅛	2¼	2
K33	2	$4^{11}/_{16}$	⅜	4⅛	45	6	⅜	12	1¼	2½	2
K34	2½	$5^{3}/_{16}$	⅜	4⅝	36	10	⅜	14	0	2½	4
K35	3	5¾	½	$5^{3}/_{16}$	36	10	⅜	14	0	3	4
K36	4	$6^{5}/_{16}$	½	5¾	30	10	⅜	18	0	2¾	4
K37	5½	$6^{15}/_{16}$	½	6⅜	30	10	⅜	20	1⅜	2¾	6
K38	7	7⅝	½	$7^{1}/_{16}$	30	10	½	20	1¾	3½	6
K39	4½	10⅜	½	$9^{13}/_{16}$	20	18	½	24	1½	3	6
K40	7	11	½	$10^{7}/_{16}$	18	18	½	28	1¾	3½	6
K41	5	$12^{13}/_{16}$	½	12¼	15	22	½	30	0	3½	4

NOTE: "0" MEANS HOLE IS ON CENTERLINE.

FIGURE 5. Typical K closure installation to duct flange.

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FIRE TEST FOR FIRE-RESISTANT VENTILATION CLOSURES

A.1. GENERAL

A.1.1 Scope. This appendix covers the requirements for testing and evaluating the elevated temperature performance of soft seated closures for use in ventilation systems. The minimum performance described by this appendix is for ventilation closures designed for 15 PSI working pressure to 200 °F and emergency fire-resistance to 1500 °F.

This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.1.2 Basis of fire test. This test procedure is intended to simulate the conditions a soft-seated closure for ventilation service may be subjected to during a fire in a ship's compartment or similar location. (see Figure A-1)

General steps to be taken are:

a. The closure shall be closed and pressure tested for through leakage in the recommended pressure direction at ambient temperature. The closure shall be shown to be leak-free using air pressure at four inches of water. The downstream side of the closure shall be open to the atmosphere with open flame protector as per Detail A.

b. The closure shall be opened and subjected to the fire for five minutes with zero differential pressure across the disc.

c. The closure shall be closed and pressurized with air to four inches of water and heated for an additional 25 minutes ensuring that the minimum body metal temperature of 800 °F is achieved. The test duration of 30 minutes total has been established on the basis that it represents the maximum time required to extinguish most shipboard fires. Fires of greater duration are to be considered of a magnitude greater than that represented by this test.

d. Airflow leakage rates shall be recorded as specified (see 6.2) at five-minute intervals during the fire test.

e. After the fire test, final qualification airflow leakage rates shall be recorded as specified (see 6.2) with the closure body between 50 °F and 100 °F and pressurized using air at four inches of water.

Note the total air leakage from all possible leak paths shall be considered part of this test including seat, flange, and stem leakage. All joints and gaskets shall be production pieces and shall not be modified or adjusted during the test. The maximum leakage is 2 $ft^3/min/ft^2$.

A.1.3 Test conditions. (see Figure A-1)

a. The closure shall not be protected with insulating material in any way during the test.

b. The closure shall be tested with the stem vertical and the bore centerline horizontal.

c. The closure shall be engulfed in a flame or flames located to the right of the vertical centerline as shown in Detail B. The number of burners used shall be sufficient to heat both neck thermocouples to 800 °F completely engulfing one side of the closure in flames before test is completed.

d. Neck thermocouples shall be recessed into the body metal $^1/_{16}$ inch and located on the closure side that is opposite the flame to prevent readings being affected by direct flame exposure.

e. Exposure time for large closures shall be increased until the minimum required neck temperature of 800 °F has been achieved.

f. Flame burner temperature shall be reduced only after neck thermocouples have reached 800 °F. The temperature at the neck thermocouples shall not drop below 800 °F for the remainder of the test time.

g. All testing shall be done with equipment and systems designed for operator safety. A pressure relief valve on the inlet piping shall be installed. No shutoff valves between the pressure relief valve and test closure shall be acceptable.

h. Once the test has begun, no adjustments shall be made to the test closure or flanges.

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i. Test pressure falling below 75% of set pressure at any time during the test shall be cause for rejection.

A.2. TEST APPARATUS

A.2.1 Fire test system. Detail A shows the arrangement of a typical fire test system (see Figure A-1). Other equipment may be added as required to meet the requirements of this test and ensure operator safety. Piping from the flowmeter to test closure "M" shall be sized to minimize air velocity pressure losses at maximum flow rates. The safety relief valve shall be set to relieve at no greater than 2 PSI. The testing system shall have an airflow capability rate greater than the maximum allowable leakage rate. This airflow rate shall be measured and recorded as specified (see 6.2) prior to the fire test with test closure "M" installed in the recommended pressure direction and the disc in the open position.

A.3. TEST PROCEDURE

A.3.1 Pre-burn. Ensure that test closure "M" is installed properly in the recommended pressure direction and that all instrumentation is installed and functioning properly. (see Figure A-1)

a. Open air supply shutoff valve "B" and set air pressure using regulator "R" to four inches of water.
b. Open test closure "M" and measure airflow to ensure conformance to A.2.1.
c. Close test closure "M" and record system pressure and airflow rate as specified (see 6.2). Check all components for zero leakage. Any leakage through the test closure "M" shall be cause for rejection.
d. Shut main air supply valve "B". Open test closure "M".

A.3.2 Burn. Record the following values as specified (see 6.2) at five-minute intervals throughout the test (see Figure A-1):

a. Time
b. Flow indicator readings
c. Temperature indicator readings
d. Test pressure gauge readings
   (1) Ignite the burner(s) and adjust the flame(s) to 1500 °F minimum at all flame thermocouples.
   (2) After five minutes of burn time, close test closure "M" and open shutoff valve "B".
   (3) Monitor all instruments and adjust flame(s) after neck thermocouples reach 800 °F to maintain 800 degrees at the necks.
   (4) After 30 minutes of total burn time, turn off the burner(s). Note that requirements A.1.3.c and A.1.3.e herein shall be met before ending the burn time.

A.3.3 Post burn. Test closure "M" shall be water sprayed to facilitate cooling immediately after the burners are turned off to simulate a fire hose (see Figure A-1). Sufficient water shall be used to cool test closure "M" to decrease neck thermocouple temperatures to between 50 °F and 100 °F within 20 minutes. The water rate shall remain constant throughout the 20 minute cooling period. Record all instrument readings, including the final qualification airflow leakage rate as specified (see 6.2).

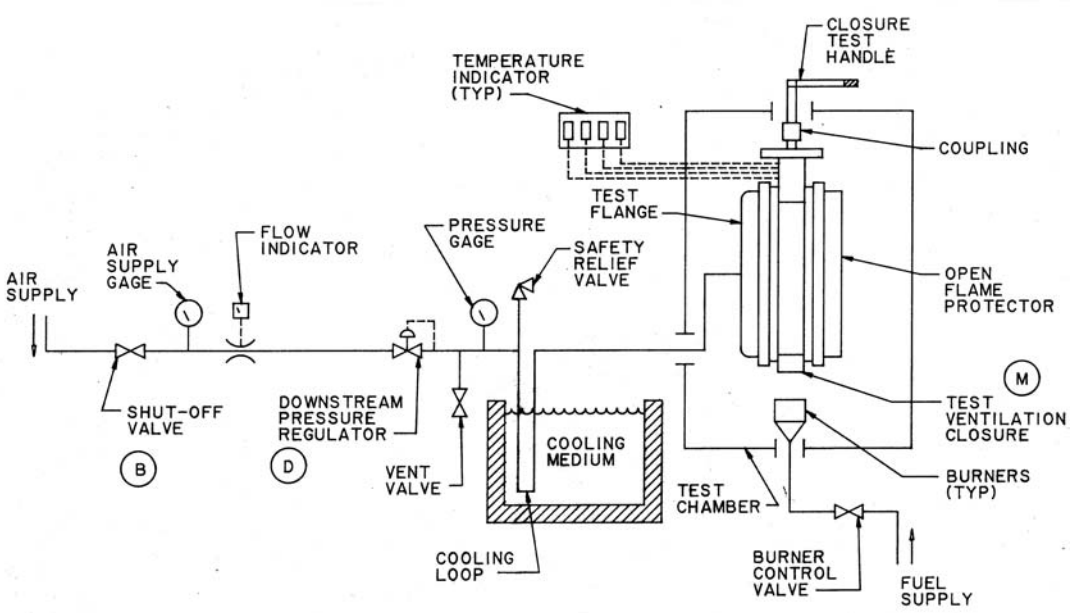
A.4. PERFORMANCE REQUIREMENTS

A.4.1 Pre-burn leakage. The test closure shall be leak-free using air at a pressure of four inches of water. The flowmeter shall register zero scfm leakage and no leakage shall be apparent by sight or sound. A leak detecting fluid (i.e., soapy water solution) shall be used on all gasket, shaft, and seat seals to ensure no leakage.

A.4.2 Burn and post burn leakage. The maximum airflow leakage rate shall not exceed 2 $ft^3/min/ft^2$.

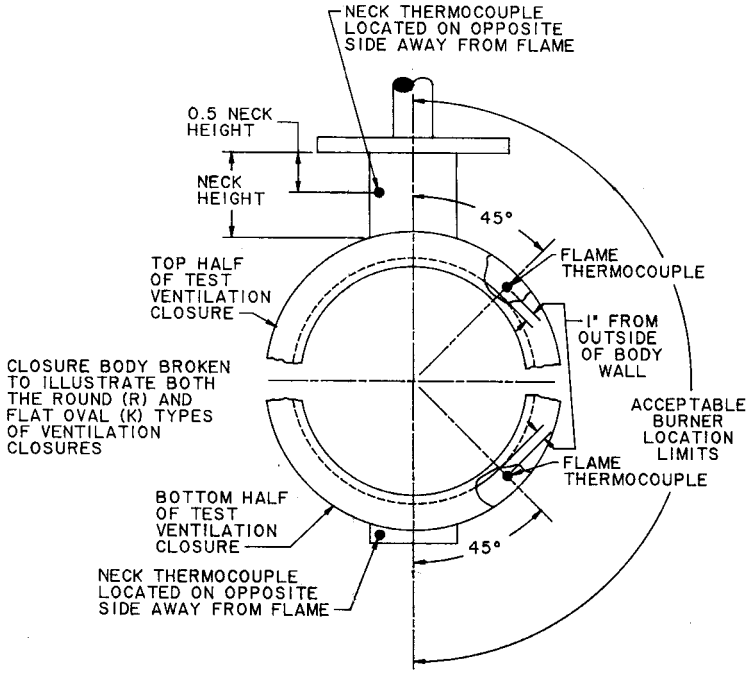
A.4.3 Acceptance by similarity. Acceptance of a closure size qualifies the size that was tested to and passed the requirements tested and all sizes below the size tested, provided the design and materials of construction are similar.

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DETAIL A

TYPICAL TEST SYSTEM



DETAIL B

THERMOCOUPLE LOCATION

FIGURE A-1. Fire test.

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